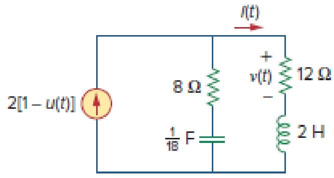
Chapter 16, Problem 52P

Problem

Given the circuit shown in Fig. 16.75, determine the values for $i(t)$ and $v(t)$ for all $t > 0$.

Figure 16.75:



Step-by-step solution

Step 1 of 6

Refer to circuit diagram in Figure 16.75 in the textbook.

For $t<0$, the circuit is in the steady state. Thus, the capacitor behave as an open circuit and the inductor behave as a short circuit.

The input current is $2\left[1-u(t)\right]\text{A}$.

It is known that,

$$u(t)=\begin{cases}1, & t>0\\ 0, & t<0\end{cases}$$

For $t<0$, the value of input current is 2A.

Redraw the circuit for $t<0$ as shown in Figure 1.

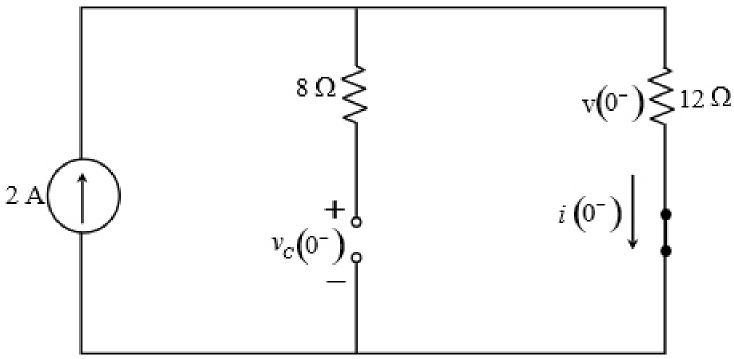


Figure 1

Step 2 of 6

Calculate the initial current through the inductor.

$$i(0^-) = 2\text{ A}$$

Calculate the voltage across $12\ \Omega$ resistor.

$$\begin{aligned} v(0^-) &= 12i(0^-) \\ &= (12)(2) \\ &= 24\text{ V} \end{aligned}$$

Since the capacitor branch is open, the voltage across the capacitor is equal to the voltage $v(0^-)$, that is,

$$\begin{aligned} v_C(0^-) &= v(0^-) \\ &= 24\text{ V} \end{aligned}$$

Step 3 of 6

For $t > 0$, calculate the value of input current.

$$\begin{aligned} I(t) &= 2[1 - u(t)] \\ &= 2(1-1) \\ &= 0\text{ A} \end{aligned}$$

Since the value of input current is 0 A for $t > 0$, the input branch is open.

Redraw the circuit for $t > 0$ in s-domain as shown in Figure 2.

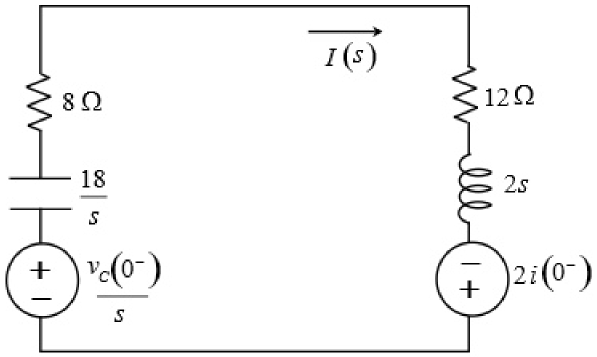


Figure 2

Step 4 of 6

Apply Kirchhoff's voltage law in the loop.

$$-\frac{v_C(0^-)}{s} + I(s)\left(\frac{18}{s} + 8 + 12 + 2s\right) - 2i(0^-) = 0$$

Substitute 24 V for $v_C(0^-)$ and 2 A for $i(0^-)$.

$$I(s)\left(2s + 20 + \frac{18}{s}\right) = \frac{24}{s} + 2(2)$$

$$I(s)\left(\frac{2s^2 + 20s + 18}{s}\right) = \frac{24 + 4s}{s}$$

$$\begin{aligned} I(s) &= \frac{2s + 12}{s^2 + 10s + 9} \\ &= \frac{2(s+6)}{(s+1)(s+9)} \end{aligned}$$

Step 5 of 6

Convert $\frac{s+6}{(s+9)(s+1)}$ into its partial fraction form.

$$\frac{s+6}{(s+9)(s+1)} = \frac{A}{s+9} + \frac{B}{s+1}$$
$$= \frac{A(s+1)+B(s+9)}{(s+9)(s+1)}$$

Compare the coefficients of s and the constant terms on both the sides.

$A+B=1$ ...... (1)

$A+9B=6$ ...... (2)

Solve equation (1) and equation (2).

$A=0.375$

$B=0.625$

The partial fraction form of $\frac{s+6}{(s+9)(s+1)}$ is $\frac{0.375}{s+9}+\frac{0.625}{s+1}$.

Step 6 of 6

The current expression is,

$$I(s) = 2\left(\frac{0.375}{s+9}+\frac{0.625}{s+1}\right)$$
$$= \frac{0.75}{s+9}+\frac{1.25}{s+1}$$

Apply inverse Laplace transform.

$$i(t) = \mathrm{L}^{-1}\left[\frac{0.75}{s+9}\right] + \mathrm{L}^{-1}\left[\frac{1.25}{s+1}\right]$$
$$= 0.75e^{-9t}u(t) + 1.25e^{-t}u(t)$$
$$= \left[0.75e^{-9t} + 1.25e^{-t}\right]u(t) \text{ A}$$

Therefore, the value of $i(t)$ is $\boxed{\left[0.75e^{-9t} + 1.25e^{-t}\right]u(t) \text{ A}}$.